

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
Department of Computer Science and Engineering

**ASSESSMENT TOOL 1 - PSEUDOCODE WRITING TASK**

|                  |                                                                                                                     |               |                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------|---------------|---------------------------------------------|
| Course Code:     | CSA0305                                                                                                             | Course Title: | Data Structures                             |
| CO Assessed:     | CO5 – Naturalization (BL4, BL6)                                                                                     | Assessment:   | Assessment Tool 1 – Pseudocode Writing Task |
| Weightage:       | 35% of CIA                                                                                                          | Total Marks:  | 100                                         |
| CO5 Statement:   | <i>Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.</i> |               |                                             |
| Student Name:    |                                                                                                                     | Reg. No.:     |                                             |
| Section / Batch: |                                                                                                                     | Date:         | 31/08/2026                                  |

**Code of Conduct**

I \_\_\_\_\_ (Name / Reg No)  
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: \_\_\_\_\_

**Instructions**

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

| Section / Topic         | Focus                         | Q Nos | Marks     |
|-------------------------|-------------------------------|-------|-----------|
| Minimum Spanning Tree   | Kruskal's Algorithm           | 1     | 20        |
| Graph Sorting           | Topological Sort              | 2     | 20        |
| Shortest Path Algorithm | MST & Dijkstra's Algorithm    | 3     | 20        |
| Shortest Path Algorithm | Dijkstra's Algorithm          | 4     | 20        |
| Minimum Spanning Tree   | Prim's or Kruskal's Algorithm | 5     | 20        |
| GRAND TOTAL:            |                               |       | 100 Marks |

## Rubrics for Calculation

| Criteria                              | Weightage | Level 4 – Exemplary                                                           | Level 3 – Proficient                                 | Level 2 – Developing                            | Level 1 – Beginning               |
|---------------------------------------|-----------|-------------------------------------------------------------------------------|------------------------------------------------------|-------------------------------------------------|-----------------------------------|
| <b>Problem Understanding</b>          | <b>20</b> | Demonstrates complete and clear understanding; handles edge cases effectively | Good understanding; minor gaps in edge case handling | Partial understanding; misses some requirements | Misinterprets problem; major gaps |
| <b>Code Correctness</b>               | <b>30</b> | Fully correct; passes all test cases                                          | Mostly correct; minor errors                         | Partially correct; several test cases fail      | Incorrect or non-functional code  |
| <b>Code Quality &amp; Readability</b> | <b>20</b> | Clean, modular, well-commented, follows standards                             | Readable with some structure                         | Limited readability; poor structure             | Unorganized and hard to read      |
| <b>Complexity Analysis</b>            | <b>20</b> | Accurate time & space complexity with justification                           | Correct but limited explanation                      | Incomplete or partially correct                 | Missing or incorrect              |
| <b>Testing &amp; Validation</b>       | <b>10</b> | Thorough testing including edge cases                                         | Adequate testing with few edge cases                 | Minimal testing                                 | No testing evidence               |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |



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DS PROGRAMMING



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### PSEUDOCODE WRITING TASK

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#### QUESTIONS

Q1

Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks



Q2

Graph - Topological Sort  
20 marks



Q3

Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's  
20 marks



Q4

Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks



Q5

Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks



#### Q1 Minimum Spanning Tree - Kruskal's Algorithm

20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).

Constraints:

- All zones must be connected
- Total installation cost must be minimized
- No redundant connections (no cycles) allowed
- The network must remain fully connected

#### Your Answer

KRUSKAL\_MST(G):

Input: Graph  $G(V, E)$  with edge weights  $w(u, v)$

Output: MST set of edges  $T$

$T \leftarrow$  empty set

sort all edges  $E$  in non-decreasing order of weight

Save Answer

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5/5 answered



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QUESTIONS

- Q1

Minimum Spanning Tree - Kruskal's Algorithm

20 marks
- Q2

Graph - Topological Sort

20 marks
- Q3

Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's

20 marks
- Q4

Shortest Path Algorithm - Dijkstra's Algorithm

20 marks
- Q5

Minimum Spanning Tree - Prim's or Kruskal's Algorithm

20 marks

Q2 Graph - Topological Sort

20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.

Constraints:

- a. Detect cycle before sorting
- b. Use DFS approach

Your Answer

TopologicalSort\_DFS(Graph):

Input: Graph with V vertices and edges (u -> v)

Output: Topological order of vertices, or report cycle

create state[V] // 0 = unvisited, 1 = visiting, 2 = visited  
create stack/order list result

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#### QUESTIONS

**Q1** Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

**Q2** Graph - Topological Sort  
20 marks

**Q3** Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's  
20 marks

**Q4** Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

**Q5** Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

#### Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's

20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.

Constraints:

- a. Use MST for infrastructure
- b. Use shortest path for routing

#### Your Answer

```
// Smart Transportation System
// Constraint a: Use MST for infrastructure (build minimal connectivity network)
// Constraint b: Use shortest path for routing (choose least-cost route per query)
```

PROCEDURE SmartTransportMST\_Dijkstra(Graph  $G(V, E)$ , CostInfra  $w$ , Source  $s$ , Destination  $d$ )

Save Answer

Next →

5/5 answered



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#### QUESTIONS

**Q1** Minimum Spanning Tree -  
Kruskal's Algorithm  
20 marks

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20 marks

**Q3** Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's  
20 marks

**Q4** Shortest Path Algorithm -  
Dijkstra's Algorithm  
20 marks

**Q5** Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm  
20 marks

#### Q4 Shortest Path Algorithm - Dijkstra's Algorithm

20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

#### Your Answer

```
// Problem: Find shortest path from S to D using a "toll-free" voucher
// Voucher use rule: exactly one edge traversal can be made free (set its weight to 0).
// We compute minimum distance where voucher can be used 0 or 1 time.
```

Algorithm TollFreeDijkstra(G, S, D):

```
// G is a graph with non-negative edge weights
```

Save Answer

Next →

5/5 answered



PSEUDOCODE WRITING TASK

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QUESTIONS

- Q1**

Minimum Spanning Tree -  
Kruskal's Algorithm

20 marks
- Q2**

Graph - Topological Sort

20 marks
- Q3**

Shortest Path Algorithm -  
Minimum Spanning Tree &  
Dijkstra's

20 marks
- Q4**

Shortest Path Algorithm -  
Dijkstra's Algorithm

20 marks
- Q5**

Minimum Spanning Tree -  
Prim's or Kruskal's Algorithm

20 marks

**Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm**

20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.  
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

**Your Answer**

SecondBestMST(G):  
Input: Weighted connected graph G(V, E)  
Output: Cost of second best MST (and optionally the tree)

- 1. MST = KruskalMST(G) // or PrimMST
- 2. mstCost = cost(MST)

Save Answer